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Studierichting: Informatica
Oefeningenreeks 2F nr. 2

$$\begin{aligned}\cosh(3x) &= \cosh(x + 2x) \\ &= \cosh(x) \cdot \cosh(2x) + \sinh(x) \cdot \sinh(2x) \\ &= \cosh(x) \cdot (\cosh^2(x) + \sinh^2(x)) + \sinh(x) \cdot 2 \cdot \sinh(x) \cdot \cosh(x) \\ &= \cosh^3(x) + \cosh(x) \cdot \sinh^2(x) + 2 \cdot \sinh^2(x) \cdot \cosh(x) \\ &= \cosh(x) \cdot (\cosh^2(x) + \sinh^2(x) + 2 \sinh^2(x)) \\ &= \cosh(x) \cdot (\cosh^2(x) + 3 \sinh^2(x)) \\ &= \cosh(x) \cdot (\cosh^2(x) + 3(\cosh^2(x) - 1)) \\ &= \cosh(x) \cdot (\cosh^2(x) + 3 \cosh^2(x) - 3) \\ &= \cosh(x) \cdot (4 \cosh^2(x) - 3)\end{aligned}$$

$$\begin{aligned}\sinh(3x) &= \sinh(x + 2x) \\ &= \sinh(x) \cdot \cosh(2x) + \cosh(x) \cdot \sinh(2x) \\ &= \sinh(x) \cdot (\cosh^2(x) + \sinh^2(x)) + \cosh(x) \cdot 2 \cdot \sinh(x) \cdot \cosh(x) \\ &= \sinh(x) \cdot \cosh^2(x) + \sinh^3(x) + 2 \sinh(x) \cosh^2(x) \\ &= \sinh(x) \cdot (\cosh^2(x) + \sinh^2(x) + 2 \cosh^2(x)) \\ &= \sinh(x) \cdot (\sinh^2(x) + 3 \cosh^2(x)) \\ &= \sinh(x) \cdot (\sinh^2(x) + 3(1 + \sinh^2(x))) \\ &= \sinh(x) \cdot (\sinh^2(x) + 3 + 3 \sinh^2(x)) \\ &= \sinh(x) \cdot (4 \sinh^2(x) + 3)\end{aligned}$$

$$\begin{aligned}
\tanh(3x) &= \tanh(x+2x) \\
&= \frac{\tanh(x) + \tanh(2x)}{1 + \tanh(x)\tanh(2x)} \\
&= \frac{\tanh(x) + \frac{2\tanh(x)}{1 + \tanh^2(x)}}{1 + \tanh(x) \cdot \frac{2\tanh(x)}{1 + \tanh^2(x)}} \\
&= \frac{\frac{\tanh(x) + \tanh^3(x) + 2\tanh(x)}{1 + \tanh^2(x)}}{\frac{1 + \tanh^2(x) + 2\tanh^2(x)}{1 + \tanh^2(x)}} \\
&= \frac{\tanh^3(x) + 3\tanh(x)}{1 + 3\tanh^2(x)}
\end{aligned}$$

References